

Forecasting domestic waste generation during successive COVID-19 lockdowns by Bidirectional LSTM super learner neural network

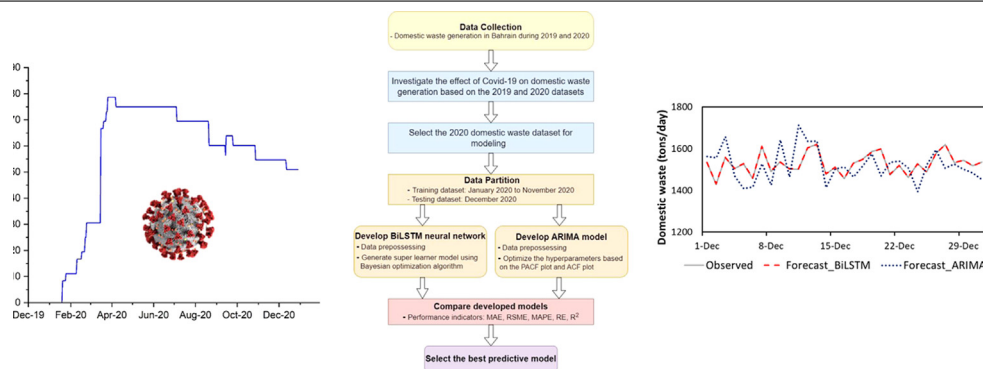
Majeed S. Jassim^{a,*}, Gulnur Coskuner^b, Nahid Sultana^c, S.M. Zakir Hossain^a

^a Department of Chemical Engineering, College of Engineering, University of Bahrain, Kingdom of Bahrain

^b Department of Civil Engineering, College of Engineering and the Built Environment, British University of Bahrain, Kingdom of Bahrain

^c Department of Computer Science, College of Computer Science and Information Technology, Imam Abdulrahman Bin Faisal University, Saudi Arabia

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 24 December 2021

Received in revised form 2 November 2022

Accepted 29 November 2022

Available online 9 December 2022

Keywords:

Solid waste

ARIMA

BiLSTM

Machine learning

Predictive modeling

ABSTRACT

Accurate prediction of domestic waste generation is a challenging task for municipalities to implement sustainable waste management strategies. In the present study, domestic waste generation in the Kingdom of Bahrain, representing a Small Island Developing State (SIDS) case study, has been investigated during successive COVID-19 lockdowns due to the pandemic in 2020. Temporal trends of daily domestic waste generation between 2019 and 2020 and their statistical analyses exhibited remarkable variations highlighting the impact of consecutive COVID-19 lockdowns on domestic waste generation. Machine learning has great potential for predicting solid waste generation rates, but only a few studies utilized deep learning approaches. The state-of-the-art Bidirectional Long Short-Term Memory (BiLSTM) network model as a deep learning method is applied to forecast daily domestic waste data in 2020. Bayesian optimization algorithm (BOA) was hybridized with BiLSTM to generate a super learner approach. The performance of the BOA-BiLSTM super learner model was further compared with the statistical ARIMA model. Performance indicators of the developed models using ARIMA and BiLSTM showed that the latter yielded superior performance for short-term forecasts of domestic waste generation. The MAE, RMSE, MAPE, and R^2 were 47.38, 60.73, 256.43, and 0.46, respectively, for the ARIMA model, compared to 3.67, 12.57, 0.24, and 0.96, respectively, for the BiLSTM model. Additionally, the relative errors for the BiLSTM model were lower than those of the ARIMA model. This study highlights that the BiLSTM can be a reliable forecasting tool for solid waste management policymakers during public health emergencies.

© 2022 Elsevier B.V. All rights reserved.

* Correspondence to: Department of Chemical Engineering, College of Engineering, University of Bahrain, Isa Town, P.O. Box 32038, Kingdom of Bahrain.

E-mail address: mjasim@uob.edu.bh (M.S. Jassim).

1. Introduction

The attainment of a sustainable society requires achieving the circular development model based upon a nexus of economic, social, and environmental dimensions. The design-driven framework of the circular economy is based upon three principles: (i) Eliminating waste and pollution, (ii) Circulating products and materials, and (iii) Regenerating nature [1]. The transformation from a linear economy to a circular one requires a commitment to implementing elements of sustainable waste management [2]. The availability of reliable historical solid waste records is essential for the proper planning of infrastructures, treatment, recovery technologies, and impact management. However, the prediction of Municipal Solid Waste (MSW) generation trends in developing countries is limited due to uncertainties and the unavailability of various factors that influence modeling techniques [3].

Artificial Intelligence (AI) algorithms include linear regression (LR) [4], time-series (TS) [5], artificial neural network (ANN) [6], supported vector machine (SVM) [7], adaptive neuro-fuzzy inference systems (ANFIS) [8], decision tree (DT) [9], gradient boosted regression (GBRT) [10], and k-nearest neighbors (k-NN) [11]. Scholarly literature applied several AI approaches to analyze and predict MSW generation. A recent review of 74 key scientific publications from the Web of Science database identified applied AI techniques to predict solid waste generation [12]. Comparatively, it was observed that 35.8% of the evaluated publications had applied ANN, 16.0% used LR, and 12.3% used SVM. Their survey showed that the application of the Autoregressive Integrated Moving Average (ARIMA) model was limited to only 1.24%.

The ANN model is extensively and successfully applied to estimate solid waste generation rates resulting high accuracy predictions [13]. However, it is susceptible to high errors originating from its black-box nature, obscuring results interpretation. Furthermore, ANN requires high accuracy of the selected predictors. The review of Abdallah et al. [13] showed that several technical, climatic, environmental, demographic, socio-economic, and legislative explanatory predictors were embedded in AI models to predict waste generation. The SVM model exhibited superior performance over ANN to predict municipal solid waste generation; however, it is highly perceptive to kernel tuning parameters [14]. As an alternative to these modeling techniques, the time-series approaches focus on the temporal distribution of solid waste generation data without explanatory variables [15]. ARIMA is a stochastic time-series methodology that predicts future values based upon the linear distribution of past observations in conjunction with random errors. The model's simplicity and systematic are its primary advantages [16]. Researchers successfully implemented the ARIMA model to forecast daily, weekly, and monthly generation rates [17,18].

Recently, the advances in computational intelligence provided momentum to apply deep learning methods for time-series prediction [19]. Those methodologies are powerful forecasting tools due to their capabilities to learn linear and non-linear relationships, robustness to noise, share uncertainties, and long-term patterns. Notably, Long Short-Term Memory (LSTM) neural network algorithm outperforms the traditional forecasting techniques such as linear and non-linear autoregressive models [20]. The effectiveness of the predictions with the LSTM approach relies heavily on the diversity and volume of historical data used in model training [21]. There are three gates in an LSTM cell: a forget gate, an input gate, and an output gate. During the training phase, LSTM can determine if there is a temporal effect on solid waste generation and how long such an effect would last with its unique three-gate structure [22].

A limited number of studies have applied LSTM neural networks as a time-series tool to forecast solid waste generation. Niu

et al. [23] used this approach, together with the ARIMA model and conventional ANN, to detect long-term effects in forecasting MSW. It was shown that LSTM neural networks are superior vis-à-vis accuracy and reliability. LSTM neural network was effectively used for forecasting weekly household waste generation rates at a single household level between 2011 and 2018 in the Municipality of Herning, Denmark [24]. Their LSTM approach improved the results by an average of 85% compared to traditional methods such as ARIMA. Furthermore, the LSTM-based forecasting model developed for construction waste data in Shanghai and Hong Kong manifested its effectiveness and accuracy [25]. However, the transmission of information for the unidirectional LSTM neural network is forward; hence, it lacks the ability to encode the reverse direction [26]. This fact is a limitation to a time-series model resulting non-capturing of key information from back to front [27,28]. Furthermore, most of the published LSTM methods for forecasting solid waste generation utilized manual search techniques to tune the hyperparameters, which is time-consuming. This may not provide the optimum results and will be difficult to perform for high-dimensional data. In this respect, several algorithms (viz., Bayesian optimization algorithm, BOA; Genetic algorithm, GA, etc.) can be integrated with each approach to tune automatically the hyperparameters of the base learner, resulting in the generation of a hybrid super learner model. Recently, a few such super learner models have been proposed utilizing small engineering datasets [29–31]. However, a more comprehensive investigative study with a wide range of variables and scopes is required.

The architecture of the Bidirectional Long Short-Term Memory (BiLSTM) network includes feedforward and feedbackward loops, consequently improving the traditional LSTM model's weakness and capturing past and future data from the time-series dataset [28]. Recent studies suggest BiLSTM algorithms exhibit superior predictions for time-series-related engineering applications [32, 33]. However, there is a lack of knowledge about its application to solid waste generation during emergencies such as pandemics.

On 30th January 2020, the World Health Organization (WHO) declared the COVID-19 outbreak as a global health emergency of international concern. Ever since the infectious pandemic spread unprecedentedly, causing millions of deaths worldwide [34]. The governmental quarantine and movement restrictions counteracted the transmission of new cases; however, these enforced policies had environmental implications, including generating various solid waste streams. COVID-19 pandemic has brought critical challenges to MSW management due to unforeseen quantity and composition variations [35].

Although the COVID-19 lockdown restrictions caused a minor waste rise in Singapore [36], a substantial MSW reduction was observed in the amount of waste reaching ten landfills/dumpsites after the lockdown in India [37]. The amount of waste increased by about 9% in Trento and 12% in Montreal due to changes in consumption behavior at the beginning of lockdown measures. Larger decreases in the quantity of commercial and construction wastes were recorded during prolonged lockdowns [38]. A current published paper identified the behavior patterns in waste collection and disposal during the first wave of COVID-19 in Regina, Saskatchewan, Canada [35]. Household waste disposal was observed to increase during the pandemic, which was attributed to working-from-home renovations. Scholarly studies focused on the amount and composition of solid waste variations during the pandemic are also associated with geographical, economic, and demographic characteristics of the investigated area.

As part of the environmental challenges encountered by the Gulf Cooperation Countries (GCC), which includes Bahrain as a member, the studies related to solid waste management are

limited. This work explores the impact of successive COVID-19 lockdowns on domestic waste generation rates and their temporal trends in Bahrain using a deep learning approach. The 2019 data was considered to represent the trends of the domestic waste generation pre-COVID-19 pandemic, while the 2020 data was accepted to reflect the trends of domestic waste generation during the successive COVID-19 lockdowns. Since the observed trend exhibited remarkable deviation between 2019 and 2020 data, the up-to-date data belonging to 2020 was utilized to construct optimal models based on a BiLSTM neural network for forecasting short-term domestic waste generation in Bahrain during the COVID-19 pandemic. Notably, the performance of the BiLSTM model depends greatly on its hyperparameters, including the number of layers, number of units in each layer, initial learning rate, and L2Regularization. However, it is necessary to tune all these hyperparameters to develop the optimized model.

The key objectives of this research can be outlined as follows: (i) Investigation of the impact of consecutive COVID-19 lockdowns on daily domestic waste generation rate in Bahrain using statistical tools, (ii) Execution of a BOA to optimize the model hyperparameters, (iii) Integrating BOA with BiLSTM to develop a novel hybrid intelligence/super learner model (e.g., BOA-BiLSTM) to robustly predict the dynamic temporal characteristics of daily domestic waste generation rate in Bahrain, (iv) Comparing the BOA-BiLSTM super learner neural network results with the statistical ARIMA model, (v) Discussing the proposed model performances using several statistical indicators (viz., MAE, RMSE, MAPE, R^2 , RE).

2. Materials and methods

2.1. Overall workflow

The first step of this study was the collection of historical data. The daily domestic waste data deposited to Askar Landfill in Bahrain has been collected from the Ministry of Municipalities Affairs and Agriculture for 2019 and 2020. In the second phase, some descriptive statistical analyses were performed to investigate whether the 2019 and 2020 datasets are significantly different from each other. Trends of daily domestic waste generation were also evaluated with time series analysis. The main steps of the overall workflow, including model development and performance evaluation, are displayed in Fig. 1.

2.2. Computational approaches

Both ARIMA and BiLSTM super learner neural networks were developed to forecast domestic waste generation. Initially, the data were partitioned into training (1 to 335) and testing (336–365) to develop both models.

2.2.1. ARIMA model

The general ARIMA model is a popular and widely applied stochastic time series model in the literature [39]. A time-series analysis uses a set of measurements taken at regular intervals to forecast values in the future for a specific activity. This method has the advantage of having good performance for either stationary or non-stationary data sets. Data in stationary time series do not depend on the time at which the series is observed. In other words, the statistical characteristics of the system, such as the mean, variance, and autocorrelation structure, do not change over time. However, there is a continuous variation in the statistical properties of the data over time in non-stationary time series. Therefore, ARIMA can be utilized to forecast both trends and seasonality variations. ARIMA is a class of models describing a time series based on past values. It denotes a linear regression

model in which the model's lags are used as predictors. It takes advantage of its delays and lagged prediction mistakes to forecast future values.

ARIMA model is characterized by three terms:

- i. p-term: it is the order of the AR term, which stands for AutoRegressive.
- ii. q-term: it is the order of the MA term, which is the Moving Average model.
- iii. d-term: it is the number of differences required to make the time series stationary.

The regression model for AR, which predicts using the past values, is given in Eq. (1).

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \dots + \beta_p Y_{t-p} + e_t \quad (1)$$

$$Y_t = f(Y_{t-1}, Y_{t-2}, Y_{t-3}, \dots, e_t) \quad (2)$$

Y_t is a function of the 'lags of Y_t ', which are Y_{t-1} , Y_{t-2} , and so on, which are the past values. $\beta_1 Y_{t-1}$ is the lag 1 of the series $\beta_2 Y_{t-2}$ is lag 2, and so on. Where α is the intercept, which is estimated by the model, and e_t is the error term.

The MA model uses the error term to predict. This model is independent of Y , so here, instead of the past values of Y , it is the error term. The model equation is mentioned below.

$$Y_t = \alpha + e_t + \phi_1 e_{t-1} + \phi_2 e_{t-2} + \dots + \phi_q e_{t-q} \quad (3)$$

$$Y_t = f(e_t, e_{t-1}, e_{t-2}, e_{t-3}, \dots) \quad (4)$$

Y_t depends only on the lagged forecast errors, represented by e_t . Where e_{t-1} and e_{t-2} represent the error. Hence Y_t is dependent on the lagged forecast errors.

The performance of the ARIMA model is dependent on its hyperparameters, such as autoregressive (AR), moving average (MA), and differencing (d). Therefore, tuning of these hyperparameters is necessary to get an optimized model. The sample partial autocorrelation function (PACF) and autocorrelation function (ACF) plots were used as a rule of thumb to determine the optimum values of these parameters.

2.2.2. Bidirectional Long Short-Term Memory (BiLSTM) neural network

LSTM network is a particular type of Recurrent Neural Network (RNN) over the standard RNN structure, and the main advantage of the LSTM network comes from its capability to learn long-term dependencies. While only one threshold function exists in the standard RNN cell structure, there are three important threshold functions in the hidden-layer cell structure of an LSTM network [40]. Information storage and interaction in the forward-propagation process are controlled by these three-threshold functions: the input, the forget, and the output gate, noted as i , f , and o , respectively. This gating mechanism resolves the gradient vanishing problem [41]. The input gate (i_t) controls the information-input process at the current time point, while the forget gate (f_t) controls the unit state information at the previous time point that needs to be discarded. Finally, the output gate (o_t) controls the filtered output of the current unit state. The following equations describe the functionality of LSTM gates [42].

$$i_t = \sigma(W_i[x_t, h_t] + b_i) \quad (5)$$

$$f_t = \sigma(W_f[x_t, h_{t-1}] + b_f) \quad (6)$$

$$o_t = \sigma(W_o[x_t, h_{t-1}] + b_o) \quad (7)$$

$$\hat{C}_t = \tanh(W_c x_t + W_c h_{t-1} + b_c) \quad (8)$$

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t \quad (9)$$

$$h_t = \tanh(C_t) * o_t \quad (10)$$

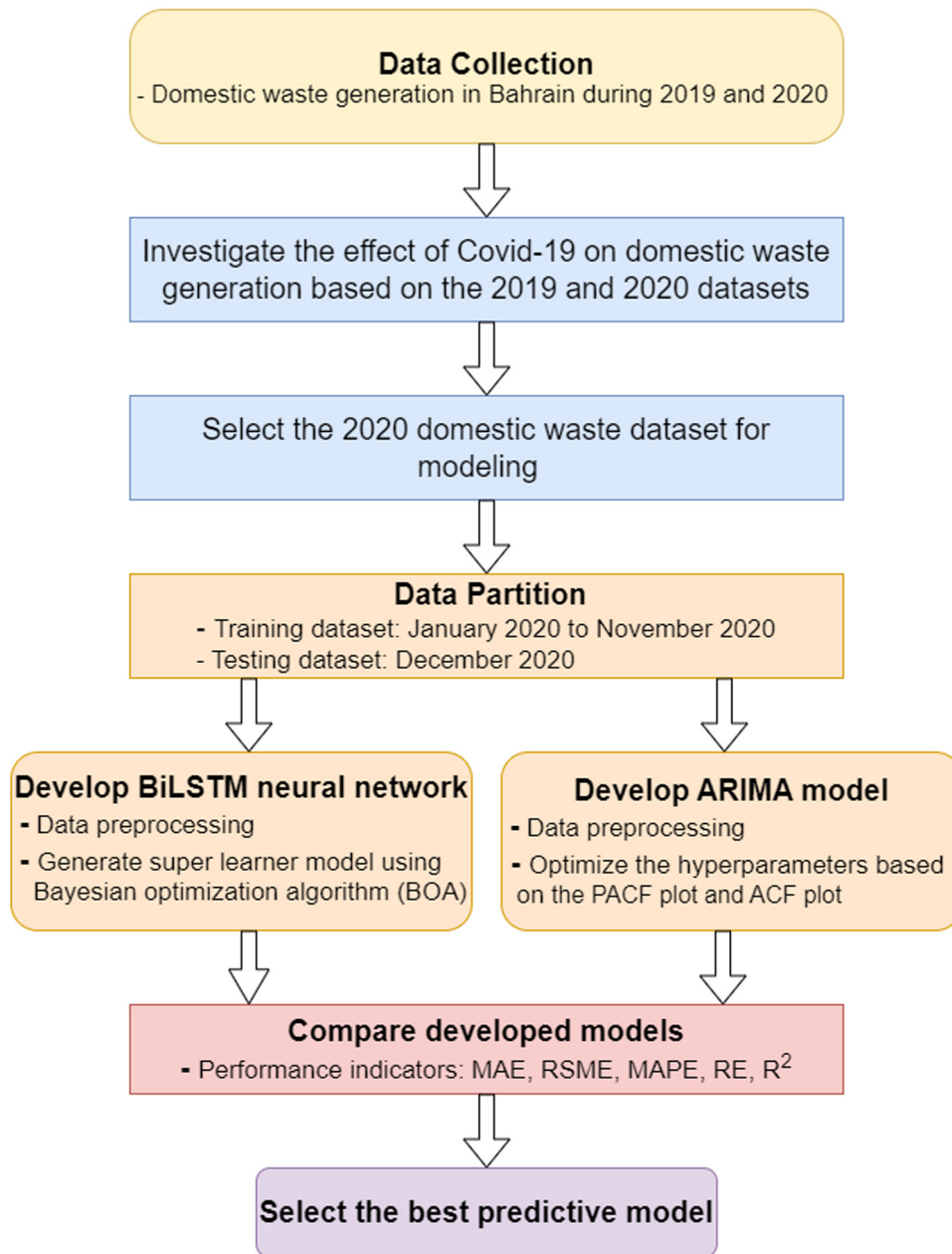


Fig. 1. The overall workflow of the present study.

where:

σ : is sigmoid activation function,

W : is the weight matrix,

x_t : is the input data at time t ,

h_t : is the hidden state at time t ,

b_i, b_o, b_f : biases,

h_{t-1} : is the prior hidden layer,

$\hat{C}$: is cell candidate state,

C_{t-1} : is the cell memory state at time $t - 1$,

$\tanh$: is the activation function.

For the Bi-LSTM network, a new forward hidden layer ($h_t^{forward}$) and backward hidden layer ($h_t^{reverse}$) are included to represent the transmission of forward and backward information, as shown in Eqs. (11)–(12). The prediction (y_t) is then calculated using Eq. (13).

$$h_t^{forward} = LSTM_t^{forward}(h_{t-1}, x_t, C_{t-1}) \quad (11)$$

$$h_t^{reverse} = LSTM_t^{reverse}(h_{t+1}, x_t, C_{t+1}) \quad (12)$$

$$y_t = [h_t^{forward}, h_t^{reverse}] \quad (13)$$

The complete dataset was standardized prior to analysis, where the values were centered around the mean with a unit standard deviation. The dropout layer works during the training process against overfitting with a given possibility of p (0.5 in this study). Training function ‘adam’ was selected based on a trial-and-error procedure. The structure of BiLSTM models was optimized by adjusting the hyperparameters such as the number of BiLSTM layers, number of units in each BiLSTM layer, initial learning rate, and L2Regularization. Several well-known tuning processes, such as manual, grid search algorithm, random search algorithm, particle swarm optimization algorithm, gravitational search algorithm, and Bayesian Optimization Algorithm (BOA), are available for tuning the hyperparameters [32]. Manual search requires users to have more background knowledge and professional expertise. It is not feasible to be applied by non-expert users. The grid search algorithm needs many different trials with a longer time

for each test. Random search is based on arbitrary combinations for a range of values; however, generally, it is not efficient in a low-dimensional space. The particle swarm optimization algorithm is a popular traditional bio-inspired approach that requires a high computational time. The gravitational search algorithm is a relatively novel population-based approach; however, this process has several extra parameters that need to be tuned. Conversely, BOA identifies the best hyperparameters in significantly less time than the abovementioned approaches. Thus, BOA was hybridized with BiLSTM to automatically find the optimum values of these parameters. Some of the major advantages of this approach include (i) requiring less computational time, (ii) applying an acquisition function that helps to evaluate the following point, and (iii) employing systematic approaches to optimize that do not need derivatives [43–46]. A brief description of BOA is given below:

BOA: The goal of Bayesian Optimization Algorithm is to choose the best parameters of the model that generate the lowest validation error. The BOA optimization can be expressed as Eq. (14):

$$s^* = \arg \min_{s \in S} f(s) \quad (14)$$

where, s^* is the set of hyperparameters that generates the smallest value of the objective score, and s indicates any value of space S .

The BOA utilizes a surrogate probability model through the Bayes' rule [47]. In this context, the next values are selected based on the data of earlier iterations. Therefore, the optimum point is selected more effectively than a random option. Mathematically, this rule can be stated as below:

$$p(w|D) = \frac{p(D|w)p(w)}{p(D)} \quad (15)$$

where $p(w)$ denotes the earlier probability, $p(D)$ denotes the evidence, $p(D|w)$ implies the likelihood, and $p(w|D)$ indicates the latter probability. In this technique, the best feasible point to assess next is measured using the acquisition function, which is a function of the surrogate model. The Gaussian Process (GP) is used to model $p(w|D)$. The model is expressed utilizing the covariance kernel function (K) and the mean function (μ) as below:

$$f(x) \sim \text{GP}(\mu, K) \quad (16)$$

Overall, the following steps are used for BOA implementation:

Step 1: Fix the objective function to minimize the validation error.

Step 2: Create a surrogate probability model of the objective function for the LSTM model.

Step 3: Get the hyperparameters that perform best on the surrogate probability model.

Step 4: Utilize these hyperparameters to the actual objective function.

Step 5: Revise the surrogate model integrating the new results.

Step 6: Recap steps 3–5 until the maximum iteration is reached.

MATLAB (version R-2019a) environment was applied to develop the hybrid super learner model. A computational flow diagram for the BOA-based hybrid super learner model is given in Fig. 2.

2.3. Performance evaluation metrics

It is very important to apply multiple performance evaluation metrics to assess the model's accuracy because the model may perform well with one metric but poorly with another. Five

performance evaluation metrics are calculated using Eqs. (17)–(21) to evaluate the performance of constructed models in this study [48]:

(1) Mean Absolute Error (MAE): MAE of a data set is the average distance between each data value, and the mean describes the variation in a data set. If MAE approaches zero, it shows the high accuracy of the model.

$$\text{MAE} = \frac{1}{n} \sum |A_v - P_v| \quad (17)$$

(2) Root Mean Square Error (RMSE): RMSE is the standard deviation of residual points, which measures data points' error from the regression line. RMSE values closer to zero also indicate the developed model's good performance.

$$\text{RMSE} = \sqrt{\frac{\sum (A_v - P_v)^2}{n}} \quad (18)$$

(3) Mean Absolute Percentage Error (MAPE %): MAPE measures the prediction accuracy of a forecasting method in statistics. It usually expresses the accuracy as a ratio defined by the formula:

$$\text{MAPE} (\%) = \frac{1}{n} \sum \frac{|A_v - P_v|}{|A_v|} * 100 \quad (19)$$

(4) Coefficient of Determination (R^2): R^2 represents the proportion of the variance for a dependent variable that is justified by an independent variable(s) in a regression model. If it approaches one, it shows the high accuracy of the constructed model.

$$R^2 = \left(\frac{\sum XY}{n \sigma_x \sigma_y} \right)^2 \quad (20)$$

(5) Relative Error (RE) is another important indicator for measuring model performance. A low value of RE indicates the high accuracy of the model.

$$\text{RE} = \frac{A_v - P_v}{A_v} \quad (21)$$

where:

n is the number of data points,

X is the dataset of the dependent variable,

Y is the dataset of explanatory variables,

σ_x is the standard deviation of dataset x ,

σ_y is the standard deviation of dataset y ,

A_v is the actual value of the data point, and

P_v is the predicted value of the data point.

3. Results and discussion

3.1. Domestic waste profile

Domestic waste generation has increased over the years as a consequence of population growth, urbanization, and economic development in Bahrain [49]. The total domestic waste deposited at Askar Landfill was 554 070.39 tons in 2019, and it increased to 559 719.87 tons in 2020, corresponding to a 1.02% rise. However, the total MSW deposited to Askar Landfill was 1,653,752 tons in 2019, then the business-as-usual scenario behaved inversely with weight reduction to 1,389,989 tons in 2020, corresponding to a negative 15.95%. The percentage of major categories of solid waste collected from different sources exhibited pertinent differences between the years 2019 and 2020. The contribution of domestic waste to total MSW was around 33.5% in 2019, and its contribution increased to 40.3% in 2020, showing a 4.8% rise.

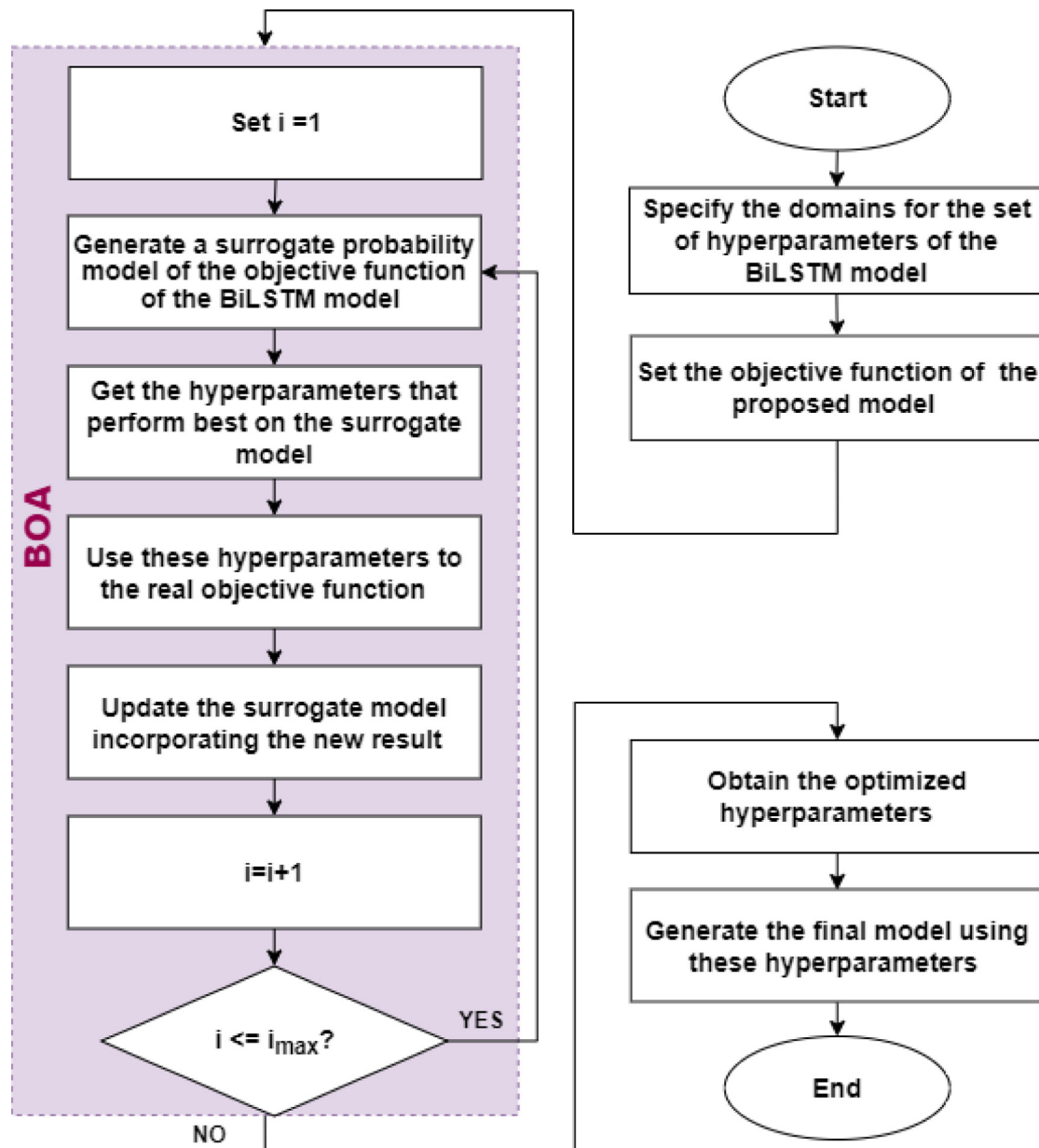


Fig. 2. Computational flow chart used for the hybrid BiLSTM and BOA.

The temporal variations of domestic waste in 2019 and 2020 were compared to elucidate the effect of successive COVID-19 lockdowns. The mean daily domestic waste generation slightly decreased from 1530.5 tons/day in 2019 to 1515.9 tons/day in 2020, representing only a 0.96% decline. However, the standard deviation of 2019 data exhibited a relatively higher ($SD = 95.1$) variation than its value in 2020 ($SD = 63.9$). The daily domestic waste generation range was 1297.7–1788.7 tons/day in 2019, while 1286.6–1722.7 tons/day in 2020. The interquartile range (IQR), the measure of statistical dispersion equal to the difference between 75th and 25th percentiles, was 131.5 for the 2019 data set compared to 81.9 for the 2020 data set.

Fig. 3(a) shows that the bell-shaped histogram of the domestic waste data in 2019 is wider and more evenly spread out than in 2020. This can also be understood from the IQR values for the datasets. Besides, this figure depicts the overall shapes of 2019 and 2020 data distributions following a normal distribution. Furthermore, the Anderson–Darling goodness-of-fit statistic (AD)

normality test was also applied, and the results show substantially smaller values of AD and the p-values (<0.005) for both datasets (see Fig. 3b), suggesting that both datasets for the years 2019 and 2020 are normally distributed. The Pearson correlation analysis was also applied for daily 2019 and 2020 domestic waste datasets. Test results showed that the computed Pearson correlation value was -0.010 and the P-value was 0.851 . This infers no significant influence of domestic waste data in 2019 on the year after (2020).

The mean daily domestic waste in 2020 was 14.6 tons more than the mean in 2019. Although the difference in the two data sets does not seem too high, the 2-sample unpaired T-test was applied to discern if this difference in the daily mean values of domestic waste is significant or not. The T-test result indicates a significant difference between these two unpaired groups of datasets ($P\text{-Value} = 0.015 < 0.05$). The variations in the amount of domestic waste generation rates between 2019 and 2020 were most probably caused by successive COVID-19 lockdowns during the pandemic in 2020.

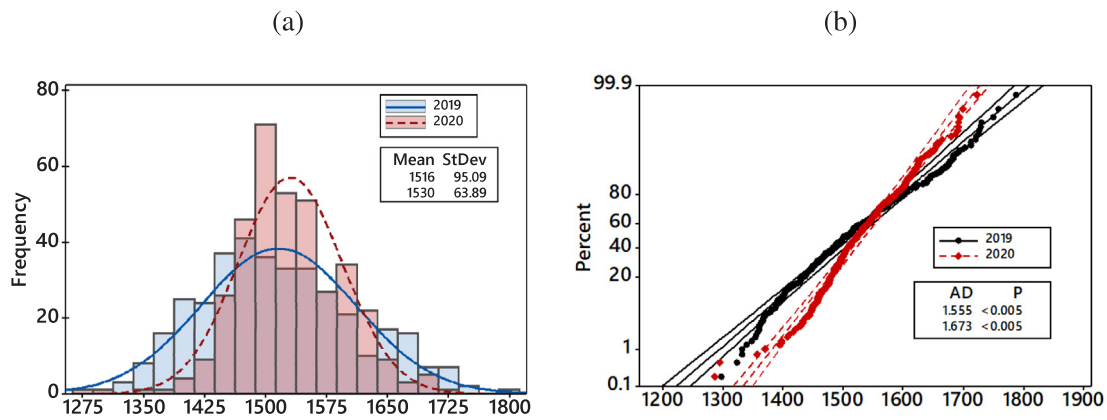


Fig. 3. (a) Histograms and (b) Probability plots of the daily domestic waste data in 2019 and 2020.

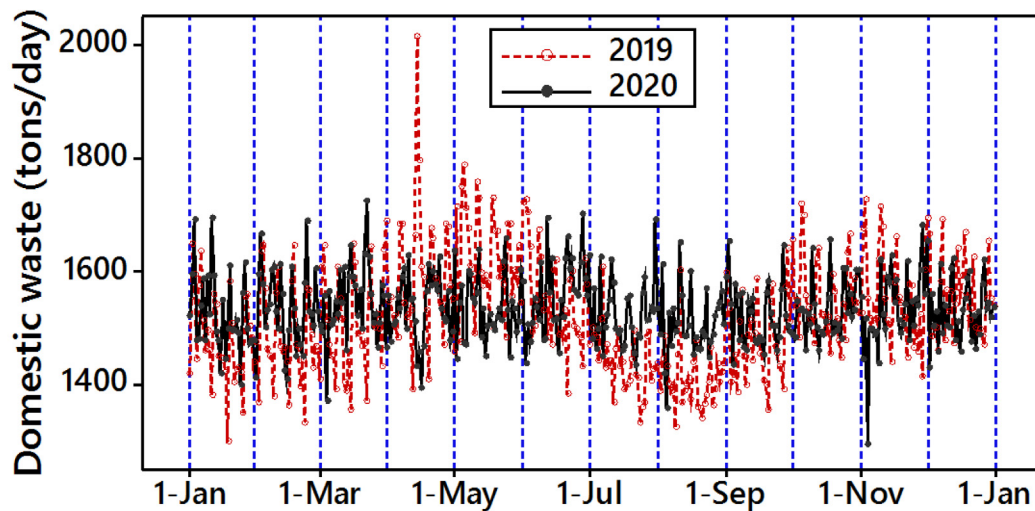


Fig. 4. Comparison of daily temporal trends of domestic waste in 2019 and 2020.

3.2. Temporal variations of domestic waste generation

Daily domestic waste data for the years 2019 and 2020 have been plotted to see the temporal variations of the dataset. Fig. 4 shows a clear seasonal variation in domestic waste generation in 2019, while an overall consistent pattern was observed in domestic waste generation in 2020. The discrepancy was unperceived in the first two months (January and February). The substantial variation started in March, while a noticeably high variation was observed from April to September. This change in the overall pattern may be due to the successive COVID-19 lockdowns that implemented various governmental restrictions.

The first case of Coronavirus was confirmed in Bahrain on 21st February 2020, when results of the COVID-19 test were positive from a bus driver that arrived from abroad. The government took quick precautionary actions and started testing people on arrival at the Bahrain International Airport. Since the count of confirmed cases increased to a total of 17 on 25 February 2020, the Bahrain government took a wide range of measures in response to the COVID-19 outbreak since 25 February 2020. The immediate measure was suspending all schools, nurseries, and universities for two weeks to avoid spreading infection. All educational institutions were closed for an additional two weeks on 05 March 2020. The Ministry of Education reported that teachers would begin working from home on 02 April 2020. Re-opening of public

schools was postponed by two weeks on 05 September 2020 (www.moh.gov.bh).

Stringency Index is a composite measure based on nine response indicators, including school closures, workplace closures, and travel bans, rescaled to a value from 0 to 100 (100 = strictest). The index records the strictness of 'lockdown style' policies that primarily restrict people's behavior. This index aims to systematically track and compare policy responses worldwide, rigorously and consistently [50]. The variation of stringency index values in Bahrain during successive COVID-19 lockdowns in 2020 are presented in Supplementary Figure S1. For the period end of March to the end of September 2020, the Stringency Index values were between 60 to 80, reflecting high restrictions on human mobility, gathering, and traveling in Bahrain. However, these restrictions were slightly reduced from October to the end of December 2020 as the number of new COVID-19 cases subsided. Due to restrictions implemented during successive COVID-19 lockdowns and changes reflected in lifestyle behavior, it was observed that seasonal variations on daily domestic waste generation are almost diminished, and it exhibited nearly a persistent pattern with some minor fluctuations over the mean value.

3.3. Development of BiLSTM super learner model

The BiLSTM neural network for domestic waste generation was developed using the daily historical data for 2020. These

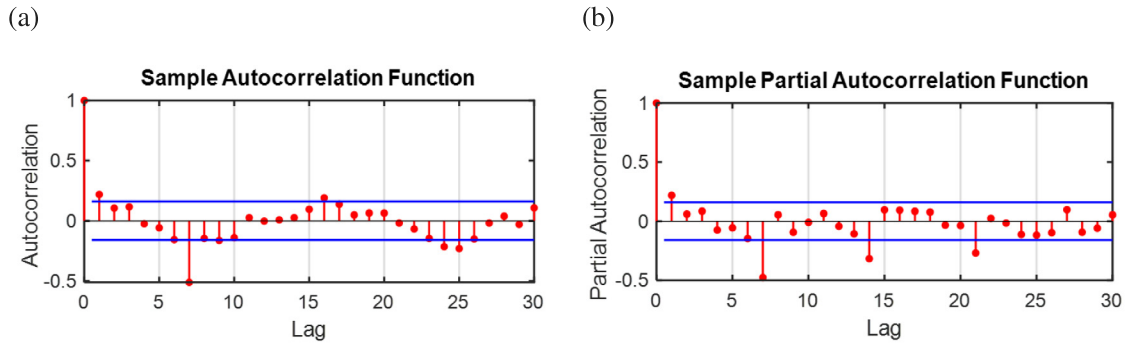


Fig. 5. (a) Sample autocorrelation function (ACF), and (b) sample partial autocorrelation function (PACF) plots of domestic waste in 2020. The Blue area denotes the confidence interval.

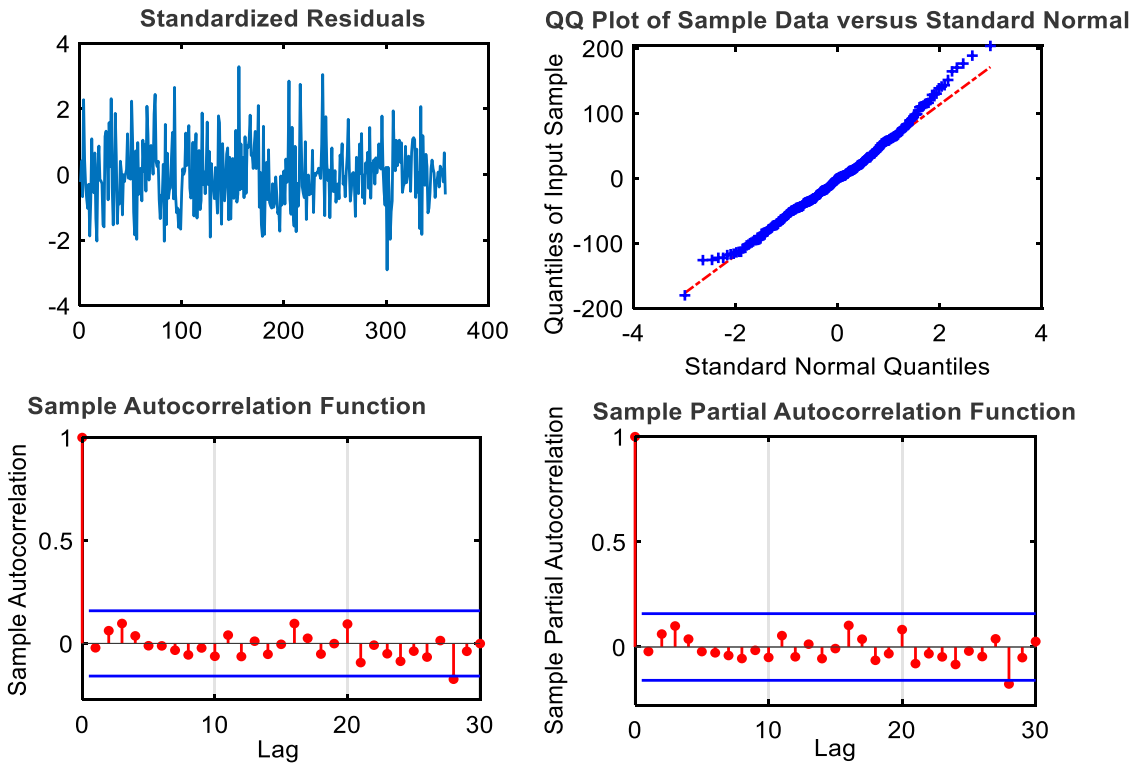


Fig. 6. Residual diagnostic plots for the developed ARIMA model.

data were allocated to training and testing datasets. The performance of the BiLSTM model depends greatly on hyperparameters (viz., number of BiLSTM layers, number of units in each BiLSTM layer, initial learning rate, L2Regularization) [51]. In this regard, the BOA approach was applied to obtain the optimal values of all hyperparameters, which provided the hybrid super learner model. The progress of the hyperparameter optimization and the feasible operating point is shown in Supplementary Figure S2. The best viable point is in a position where the observed objective score or validation error is the smallest. The validation error was found to be 0.00066 at 35 iterations. The tuned hyperparameters were achieved at this location. Table S1 presents the tuned hyperparameters used to determine the optimal super learner model. The total computational elapsed time was also found to be low (118.2 s). It is noteworthy that developing any super learner model for a particular dataset requires substantial time, research, and computational trials (experiments). Thus, finding an optimized model is a major requirement, and BOA plays a vital

role. Recently, the deep learning approach has become one of the crucial research areas across multiple engineering branches, including its application under the purview of MSW management. Hence, such a super learner-based modeling approach is not only a state-of-the-art application but also a potential area of study.

3.4. Development of the ARIMA model

The performance of the BiLSTM model was compared with the ARIMA model. In this regard, seasonal differencing (with periodicity 7) was used for the data pre-processing. It is noteworthy that several hyperparameters such as autoregressive (AR), moving average (MA), and differencing (d) control the performance of the ARIMA model. Thus, tuning of these parameters is necessary to get an optimized model. The values of AR and MA were determined based on the sample partial autocorrelation function (PACF) and autocorrelation function (ACF) plots of the domestic waste in 2020 (see Fig. 5).

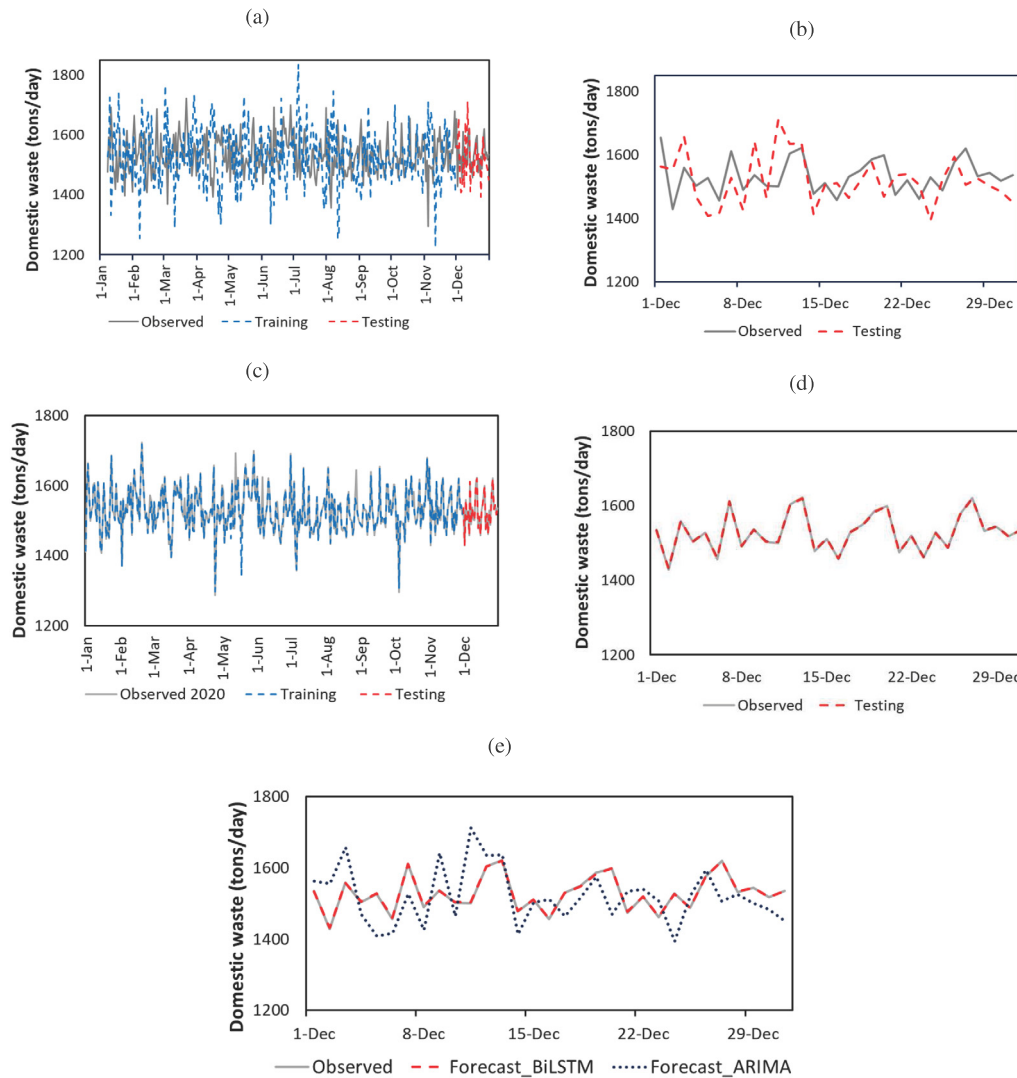


Fig. 7. Forecasting of daily domestic waste generation by using (a) ARIMA (Jan–Dec 2020); (b) ARIMA (Dec 2020); (c) BiLSTM (Jan–Dec 2020); (d) BiLSTM (Dec 2020); and (e) comparison chart of ARIMA and the proposed BiLSTM model.

Notably, the ACF explains how the current value of a time series is compared to the prior values. The correlation coefficient is represented on the x-axis, while the number of lags is represented on the y-axis. PACF offers a partial correlation between the time series and its lagged values. The main difference between them is that the PACF plot regressed the time series values at all smaller delays, while the ACF, on the other hand, does not account for additional delays.

If the values of ACF and PACF surpass the confidence interval, the lag time must be considered for parameters AR(p) and MA(q). Thus, based on the plot of ACF, the significant lags are at 1, 7, and 25, while the PACF plot indicates that the significant lags are at 1, 7, 14, 21, and 25. Based on this observation along with several experiments, the optimized value of AR was selected as 7, 14, 21, and 25, while the values of MA and d were selected as 1. These optimized values were chosen based on the Akaike Information Criterion (AIC) lowest values of 3640.6, and the Bayesian Information Criterion (BIC) of 3670.92. A noteworthy aspect is that the AIC and BIC are the criteria for the model's goodness of fit. Lower values of AIC and BIC represent a better fit for the model [52,53].

Based on the summary of the ARIMA model (see Table 1), all the AR and MA coefficients are significantly different from

Table 1
Summary of ARIMA models.

	Value	Standard error	T-statistic	P-value
Constant	−0.26635	4.1351	−0.06441	0.94864
AR{7}	−0.77501	0.052186	−14.851	6.86E−50
AR{14}	−0.53312	0.056064	−9.5092	1.92E−21
AR{21}	−0.30813	0.050678	−6.0802	1.20E−09
AR{25}	−0.16059	0.040515	−3.9637	7.38E−05
MA{1}	0.13665	0.057019	2.3966	1.65E−02
DoF	19.288	22.201	0.8688	3.85E−01
Variance	3831.5	344.52	11.121	9.88E−29

AIC: 3640.6, BIC: 3670.92.

zero, have a T-statistic greater than 2 in magnitude, and the P-values are all less than 0.05. These results indicate an appropriate selection of the hyperparameters for the model.

Residual diagnostic plots (see Fig. 6) were analyzed to check the goodness of fit of the developed ARIMA model. The quantile–quantile plot (QQ-plot) shows no obvious violations of the normality assumption. The sample autocorrelation function (ACF) and partial autocorrelation function (PACF) plots for the standardized residuals exhibit no significant autocorrelation, confirming that the residuals are uncorrelated. The first 334 observations

Table 2
Performance evaluation metrics for optimal ARIMA and BiLSTM models.

Model		Training	Testing	Overall
ARIMA	MAE	100.8709	34.7991	47.3846
	RMSE	130.7766	46.9708	60.7314
	MAPE%	342.3387	171.9111	256.4296
	R ²	0.5879	0.6688	0.458
BiLSTM	MAE	3.8756	1.3944	3.6717
	RMSE	13.1163	1.6442	12.5745
	MAPE%	0.2500	0.0918	0.2370
	R ²	0.9552	0.9989	0.9576

Table S1
Optimized BiLSTM hyperparameters obtained using BOA.

Number of Layer	Number of Units	Initial Learning Rate	L2Regularization
1	150	0.0102	1.1097e-10

were used to develop the model and then forecast the next 31 observations to verify the predictive performance.

3.5. Performance comparison of ARIMA and proposed model

The performance comparison between the historical and the forecasting data of the optimized BiLSTM super learner neural network and the ARIMA model is shown in Fig. 7. The ARIMA model (see Fig. 7(a) and (b)) does not fit well at the peak and valley of the curve, while the BiLSTM super learner neural network (see Fig. 7(c) and (d)) fits well and almost overlaps on the observed values. Fig. 7(e) illustrates the superiority of the proposed model compared to ARIMA model forecasting. Besides, the ARIMA model fails to offer a reliable prediction for most data points, while the optimized BiLSTM super learner neural network seems to perform just as well on the training dataset.

Four performance evaluation metrics known as MAE, RMSE, MAPE%, and R² are calculated for the optimal ARIMA and BiLSTM super learner models [54,55]. The values of these indicators on the training, testing, and overall datasets are displayed in Table 2. The accuracy of the ARIMA model increases from the training dataset to the testing dataset; however, the overall high errors (MAE: 47.3846, RMSE: 60.7314, MAPE%: 256.4296) and low R² (0.458) were observed. Thus, the ARIMA model fails to simulate domestic waste generation effectively. On the other hand, the BiLSTM neural network shows stable performance with low overall errors (MAE: 3.67, RMSE: 12.57, MAPE%: 0.24) and high R² (0.96). This indicates that the BiLSTM super learner model can effectively forecast domestic waste generation. This also shows that once continuous daily historical data are available for one year, the BiLSTM super learner model can accentuate robust forecasts for the near future.

Furthermore, the relative errors based on the ARIMA and BiLSTM models for the whole dataset are also shown in Figure S3. The results exhibit a very high relative error in the ARIMA model compared to the BiLSTM model. The relative error in the BiLSTM model ranges from 0 to 0.074, while in the ARIMA model, this range is from 0 to 116. The overall results indicated that the BiLSTM super learner model performs much better than the ARIMA model.

4. Conclusions

Herein, the effect of successive lockdowns during COVID-19 pandemic on domestic solid waste generation in Bahrain was investigated. Statistical analysis indicated that domestic waste generation was significantly higher in 2020 compared to 2019 due to the effect of implementations of working from home,

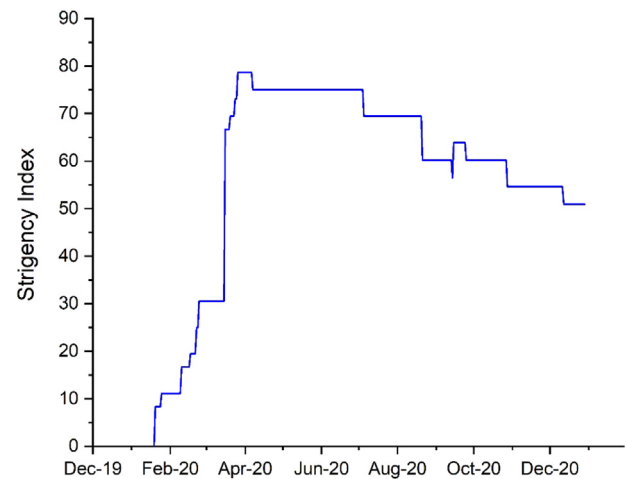


Fig. S1. Stringency Index values in Bahrain during successive COVID-19 lockdowns in 2020.

Min objective vs. Number of function evaluations

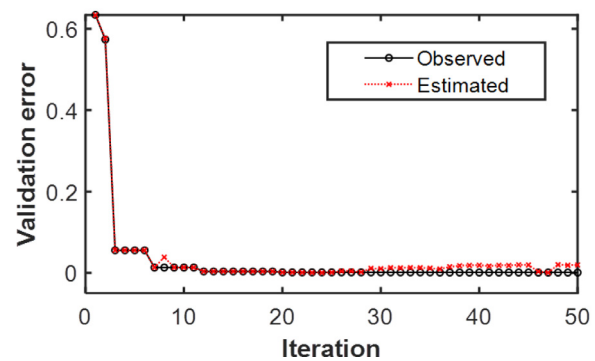


Fig. S2. The progress of Bayesian optimization for tuning hyperparameters of BiLSTM. The estimated score is the result of the surrogate probability model, while the observed score is the value of the true objective function. The observed validation error was found to be 0.00066 at 35 iterations.

online education, and restrictions on visiting restaurants, gyms, pools, and indoor facilities. Additionally, the coronaphobia has lowered human mobility and big gatherings during the whole pandemic period. Thus, applying those lockdown measures seems to promote inhabitants' consumption at their houses in Bahrain, leading to higher household waste generation throughout the year 2020.

A novel BOA-based BiLSTM super learner neural network approach was successfully developed to forecast the daily domestic waste generation in 2020. Based on the results, the performance of the BiLSTM model was superior to the conventional ARIMA model using selected evaluation metrics. It is worth mentioning that public health emergencies provide limited historical data; hence, it is important to develop reliable and economical tools for robust forecasting of waste generation rates in the short term.

Future comparative studies between pre- and during the COVID-19 period can be conducted for other major waste streams, such as commercial and construction waste generation as well as MSW generation. The time-series scale can also be extended to include the post-COVID-19 era to investigate further variations for different solid waste generation rates. Based on our key findings, we strongly propose applying this reliable modeling tool during emergencies to assist in implementing sustainable waste management policies and practices.

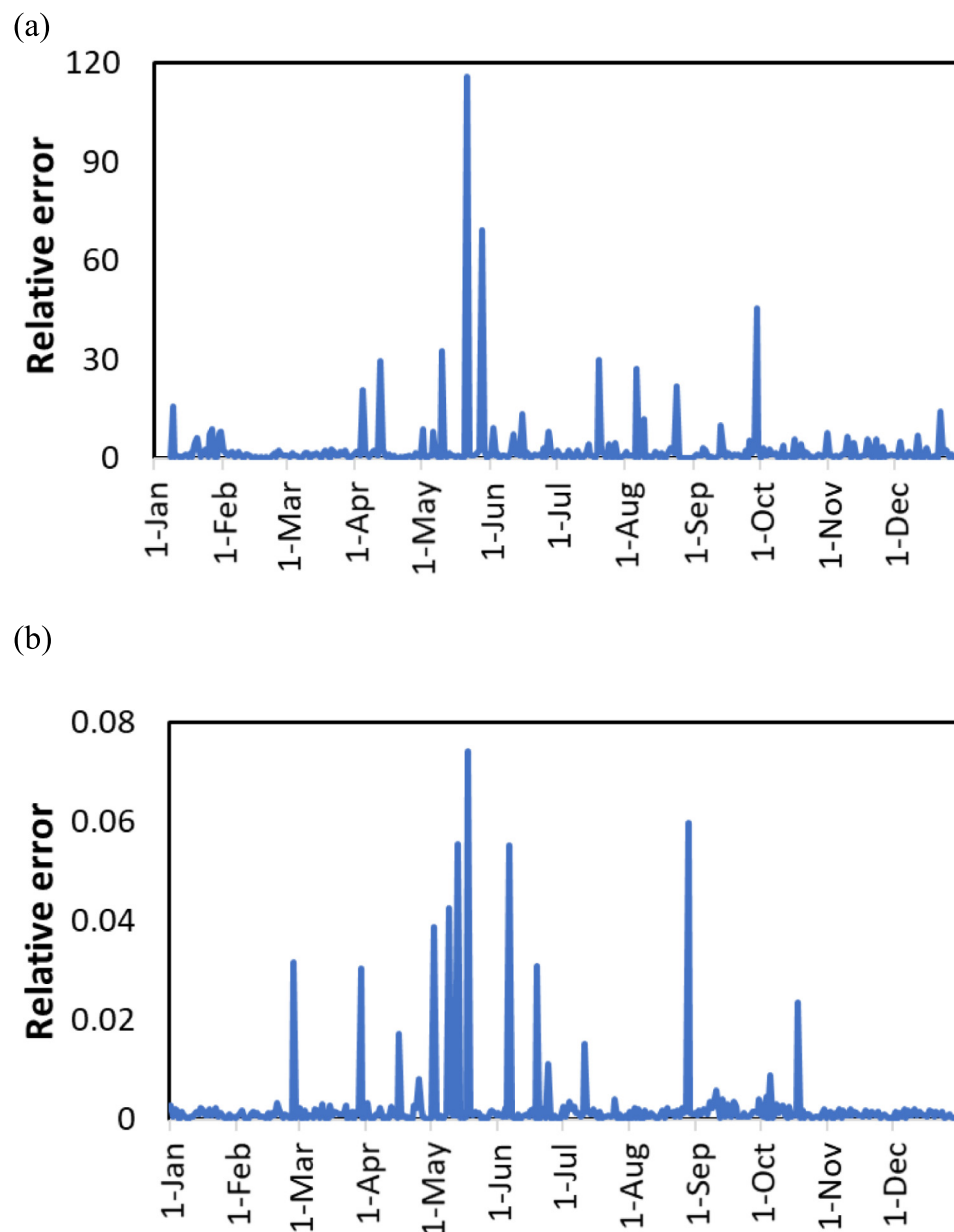


Fig. S3. Relative error for the whole dataset (a) ARIMA model and (b) BiLSTM model.

CRediT authorship contribution statement

Majeed S. Jassim: Data curation, Investigation, Methodology, Project administration, Visualization, Writing – review & editing. **Gulnur Coskuner:** Conceptualization, Data curation, Investigation, Methodology, Supervision, Visualization, Writing – original draft. **Nahid Sultana:** Formal analysis, Methodology, Software, Validation, Visualization, Writing – review & editing. **S.M. Zakir Hossain:** Methodology, Project administration, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Opinions, findings, conclusions, or recommendations expressed herein are those of the authors. They are grateful to the Ministry of Municipalities Affairs and Agriculture for providing daily domestic solid waste data for 2019 and 2020.

Appendix. Supplementary data

See Figs. S1–S3 and Table S1.

References

- [1] Ellen MacArthur Foundation, *What Is a Circular Economy? A Framework for an Economy that is Restorative and Regenerative by Design*, 2020.
- [2] J.K. Seadon, Sustainable waste management systems, *J. Clean. Prod.* 18 (2010) 1639–1651, <http://dx.doi.org/10.1016/j.jclepro.2010.07.009>.
- [3] K.A. Kolekar, T. Hazra, S.N. Chakrabarty, A review on prediction of municipal solid waste generation models, *Procedia Environ. Sci.* 35 (2016) 238–244, <http://dx.doi.org/10.1016/j.proenv.2016.07.087>.

- [4] S. Shalev-Shwartz, S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*, Cambridge University Press, 2014, <http://dx.doi.org/10.1017/CBO9781107298019>.
- [5] T.C. Fu, A review on time series data mining, *Eng. Appl. Artif. Intell.* 24 (2011) 164–181, <http://dx.doi.org/10.1016/j.engappai.2010.09.007>.
- [6] S. Haykin, *Neural Networks: A Comprehensive Foundation*, Prentice Hall, New Jersey, 1994.
- [7] V. Vapnik, *The Nature of Statistical Learning Theory*, Springer science & business media, 1999.
- [8] J.S.R. Jang, ANFIS: Adaptive-network-based fuzzy inference system, *Bull. Math. Biophys.* 5 (1943) 665–685, <http://dx.doi.org/10.1109/21.256541>.
- [9] C. Wu, *The Decision Tree Approach to Classification*, Purdue University, 1975.
- [10] L. Breiman, *Arcing the Edge*, Technical Report 486, Statistics Department, University of California at ..., 1997.
- [11] T.M. Cover, P.E. Hart, Nearest neighbor pattern classification, *IEEE Trans. Inf. Theory* 13 (1967) 21–27, <http://dx.doi.org/10.1109/TIT.1967.1053964>.
- [12] J.L. Dias, M.K. Sott, C.C. Ferrão, J.C. Furtado, J.A.R. Moraes, Data mining and knowledge discovery in databases for urban solid waste management: A scientific literature review, *Waste Manag. Res.* 39 (2021) 1331–1340, <http://dx.doi.org/10.1177/0734242X211042276>.
- [13] M. Abdallah, M. Abu Talib, S. Feroz, Q. Nasir, H. Abdalla, B. Mahfood, Artificial intelligence applications in solid waste management: A systematic research review, *Waste Manage.* 109 (2020) 231–246, <http://dx.doi.org/10.1016/j.wasman.2020.04.057>.
- [14] M.S. Jassim, G. Coskuner, M. Zontul, Comparative performance analysis of support vector regression and artificial neural network for prediction of municipal solid waste generation, *Waste Manag. Res.* 40 (2022) 195–204, <http://dx.doi.org/10.1177/0734242X211008526>.
- [15] I. Rimaityte, T. Ruzgas, G. Denafas, V. Račys, D. Martuzevicius, Application and evaluation of forecasting methods for municipal solid waste generation in an Eastern-European city, *Waste Manag. Res.* 30 (2012) 89–98, <http://dx.doi.org/10.1177/0734242X13096754>.
- [16] C. Zafra, Y. Angel, E. Torres, ARIMA analysis of the effect of land surface coverage on PM10 concentrations in a high-altitude megacity, *Atmos. Pollut. Res.* 8 (2017) 660–668, <http://dx.doi.org/10.1016/j.apr.2017.01.002>.
- [17] J. Navarro-Esbrí, E. Diamadopoulos, D. Ginestar, Time series analysis and forecasting techniques for municipal solid waste management, *Resour. Conserv. Recycl.* 35 (2002) 201–214, [http://dx.doi.org/10.1016/S0921-3449\(02\)00002-2](http://dx.doi.org/10.1016/S0921-3449(02)00002-2).
- [18] X. Fang, Z. Yuan, Performance enhancing techniques for deep learning models in time series forecasting, *Eng. Appl. Artif. Intell.* 85 (2019) 533–542, <http://dx.doi.org/10.1016/j.engappai.2019.07.011>.
- [19] C.-Y. Zhang, C.L.P. Chen, M. Gan, L. Chen, Predictive deep Boltzmann machine for multiperiod wind speed forecasting, *IEEE Trans. Sustain. Energy* 6 (2015) 1416–1425, <http://dx.doi.org/10.1109/TSTE.2015.2434387>.
- [20] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput.* 9 (1997) 1735–1780, <http://dx.doi.org/10.1162/neco.1997.9.8.1735>.
- [21] H. Shi, M. Xu, R. Li, Deep learning for household load forecasting—a novel pooling deep RNN, *IEEE Trans. Smart Grid* 9 (2018) 5271–5280, <http://dx.doi.org/10.1109/TSG.2017.2686012>.
- [22] J.K. Solano Meza, D. Orjuela Yepes, J. Rodrigo-Illarri, E. Cassiraga, Predictive analysis of urban waste generation for the city of Bogotá, Colombia, through the implementation of decision trees-based machine learning, support vector machines and artificial neural networks, *Heliyon* 5 (2019) <http://dx.doi.org/10.1016/j.heliyon.2019.e02810>.
- [23] D. Niu, F. Wu, S. Dai, S. He, B. Wu, Detection of long-term effect in forecasting municipal solid waste using a long short-term memory neural network, *J. Clean. Prod.* 290 (2021) <http://dx.doi.org/10.1016/j.jclepro.2020.125187>.
- [24] M. Cubillos, Multi-site household waste generation forecasting using a deep learning approach, *Waste Manage.* 115 (2020) 8–14, <http://dx.doi.org/10.1016/j.wasman.2020.06.046>.
- [25] L. Huang, T. Cai, Y. Zhu, Y. Zhu, W. Wang, K. Sun, Lstm-based forecasting for urban construction waste generation, *Sustain.* 12 (2020) 1–12, <http://dx.doi.org/10.3390/su12208555>.
- [26] L. Wang, X. Xu, Q. Su, Y. Song, H. Wang, M. Xie, Automatic gear shift strategy for manual transmission of mine truck based on Bi-LSTM network, *Expert Syst. Appl.* 209 (2022) 118197, <http://dx.doi.org/10.1016/j.eswa.2022.118197>.
- [27] F.A. Gers, N.N. Schraudolph, J. Schmidhuber, Learning precise timing with LSTM recurrent networks, *J. Mach. Learn. Res.* 3 (2003) 115–143, <http://dx.doi.org/10.1162/153244303768966139>.
- [28] A. Graves, J. Schmidhuber, Framewise phoneme classification with bidirectional LSTM and other neural network architectures, *Neural Netw.* 18 (2005) 602–610, <http://dx.doi.org/10.1016/j.neunet.2005.06.042>.
- [29] O. Owoyele, P. Pal, A.V. Torreira, D. Probst, M. Shaxted, M. Wilde, P.K. Senecal, Application of an automated machine learning-genetic algorithm (AutoML-GA) coupled with computational fluid dynamics simulations for rapid engine design optimization, 2021, <http://dx.doi.org/10.1177/14680874211023466>.
- [30] O. Owoyele, P. Pal, A. Vidal Torreira, An automated machine learning-genetic algorithm framework with active learning for design optimization, *J. Energy Resour. Technol.* 143 (2021) <http://dx.doi.org/10.1115/1.4050489>.
- [31] A.A. Moiz, P. Pal, D. Probst, Y. Pei, Y. Zhang, S. Som, J. Kodavasal, A machine learning-genetic algorithm (ML-GA) approach for rapid optimization using high-performance computing, *SAE Int. J. Commer. Veh.* 11 (2018) 291–306, <http://dx.doi.org/10.4271/2018-01-0190>.
- [32] J. He, J. Cui, Y. Zhao, G. Zhang, M. Xue, D. Chu, Spatial-temporal seizure detection with graph attention network and Bi-directional Lstm architecture, *SSRN Electron. J.* 78 (2022) 103908, <http://dx.doi.org/10.2139/ssrn.3987849>.
- [33] K. Feng, Z. Fan, A novel bidirectional LSTM network based on scale factor for atrial fibrillation signals classification, *Biomed. Signal Process. Control* 76 (2022) 103663, <http://dx.doi.org/10.1016/j.bspc.2022.103663>.
- [34] C. Sohrabi, Z. Alsafi, N. O'Neill, M. Khan, A. Kerwan, A. Al-Jabir, C. Iosifidis, R. Agha, World health organization declares global emergency: A review of the novel coronavirus (COVID-19), *Int. J. Surg.* 76 (2020) 2019) 71–76, <http://dx.doi.org/10.1016/j.ijsu.2020.02.034>.
- [35] H.L. Vu, K.T.W. Ng, A. Richter, N. Karimi, G. Kabir, Modeling of municipal waste disposal rates during COVID-19 using separated waste fraction models, *Sci. Total Environ.* 789 (2021) <http://dx.doi.org/10.1016/j.scitotenv.2021.148024>.
- [36] Y. Van Fan, P. Jiang, J.J. Klemeš, P.Y. Liew, C.T. Lee, Integrated regional waste management to minimise the environmental footprints in circular economy transition, *Resour. Conserv. Recycl.* 168 (2021) <http://dx.doi.org/10.1016/j.resconrec.2020.105292>.
- [37] M. Somani, A.N. Srivastava, S.K. Gummadivalli, A. Sharma, Indirect implications of COVID-19 towards sustainable environment: An investigation in Indian context, *Bioresour. Technol. Rep.* 11 (2020) <http://dx.doi.org/10.1016/j.biteb.2020.100491>.
- [38] M. Cai, C. Guy, M. Héroux, E. Lichtfouse, C. An, The impact of successive COVID-19 lockdowns on people mobility, lockdown efficiency, and municipal solid waste, *Environ. Chem. Lett.* 19 (2021) 3959–3965, <http://dx.doi.org/10.1007/s10311-021-01290-z>.
- [39] G. Box, G. Jenkins, G. Reinsel, G. Ljung, *Time Series Analysis: Forecasting and Control*, fifth ed., John Wiley & Sons, Hoboken, New Jersey, 2016.
- [40] A. Graves, Long short-term memory, in: *Supervised Seq. Label. with Recurr. Neural Networks*, in: *Stud. Comput. Intell.*, Springer, Berlin, Heidelberg, 2012.
- [41] Y. Bengio, P. Simard, P. Frasconi, Learning long-term dependencies with gradient descent is difficult, *IEEE Trans. Neural Netw.* 5 (1994) 157–166, <http://dx.doi.org/10.1109/72.7279181>.
- [42] K. Greff, R.K. Srivastava, J. Koutnik, B.R. Steunebrink, J. Schmidhuber, LSTM: A search space odyssey, *IEEE Trans. Neural Netw. Learn. Syst.* 28 (2017) 2222–2232, <http://dx.doi.org/10.1109/TNNLS.2016.2582924>.
- [43] J. Wu, X.Y. Chen, H. Zhang, L.D. Xiong, H. Lei, S.H. Deng, Hyperparameter optimization for machine learning models based on Bayesian optimization, *J. Electron. Sci. Technol.* 17 (2019) 26–40, <http://dx.doi.org/10.11989/JEST.1674-862X.80904120>.
- [44] J. Snoek, H. Larochelle, R.P. Adams, Practical Bayesian optimization of machine learning algorithms, in: *Adv. Neural Inf. Process. Syst.*, 2012, pp. 2951–2959.
- [45] M. Feurer, F. Hutter, Hyperparameter optimization, in: *Autom. Mach. Learn.*, Springer, Cham, 2019, pp. 3–33, http://dx.doi.org/10.1007/978-3-030-05318-5_1.
- [46] J. Mockus, *Global Optimization and the Bayesian Approach*, D. Reidel Publishing Company, 1989, http://dx.doi.org/10.1007/978-94-009-0909-0_1.
- [47] B. Shahriari, K. Swersky, Z. Wang, R.P. Adams, N. De Freitas, Taking the human out of the loop: A review of Bayesian optimization, *Proc. IEEE* 104 (2016) 148–175, <http://dx.doi.org/10.1109/JPROC.2015.2494218>.
- [48] S. Makridakis, S.C. Wheelwright, R.J. Hyndman, *Forecasting Methods and Applications*, John Wiley & Sons, 2008.
- [49] G. Coskuner, M.S. Jassim, N. Nazeer, G.H. Damindra, Quantification of landfill gas generation and renewable energy potential in arid countries: Case study of Bahrain, *Waste Manag. Res.* 38 (2020) 1110–1118, <http://dx.doi.org/10.1177/0734242X20933338>.
- [50] E. O-O, M.R. Edouard Mathieu, Hannah Ritchie, Lucas Rodés-Guirao, Cameron Appel, Charlie Giattino, Joe Hasell, Bobbie Macdonald, Saloni Dattani, Diana Beltekian, Coronavirus Pandemic (COVID-19), Our World Data, 2022, <https://ourworldindata.org/coronavirus>.

- [51] J. Ma, Y. Ding, J.C.P. Cheng, F. Jiang, Z. Wan, A temporal-spatial interpolation and extrapolation method based on geographic long short-term memory neural network for PM2.5, *J. Clean. Prod.* 237 (2019) 117729, <http://dx.doi.org/10.1016/j.jclepro.2019.117729>.
- [52] R. McElreath, *Statistical Rethinking. A Bayesian Course with Examples in R and Stan*, second ed., CRC Press, 2020.
- [53] M. Taddy, *Business Data Science : Combining Machine Learning and Economics to Optimize, Automate, and Accelerate Business Decisions*, McGraw-Hill, New York, 2019.
- [54] B. Cortez, B. Carrera, Y.J. Kim, J.Y. Jung, An architecture for emergency event prediction using LSTM recurrent neural networks, *Expert Syst. Appl.* 97 (2018) 315–324, <http://dx.doi.org/10.1016/j.eswa.2017.12.037>.
- [55] M.S.S. Abujazar, S. Fatihah, I.A. Ibrahim, A.E. Kabeel, S. Sharil, Productivity modelling of a developed inclined stepped solar still system based on actual performance and using a cascaded forward neural network model, *J. Clean. Prod.* 170 (2018) 147–159, <http://dx.doi.org/10.1016/j.jclepro.2017.09.092>.